

$\pi_{0.5}$ · arXiv 2504.16054 · Physical Intelligence

Scientific Reviewer

8+3 · seat 4

Strong system · citation hygiene
Hybrid FAST→FM must be accurate

Sep 30 paper club · VLA Architectures

Spine: open-world + hybrid FAST discrete pretrain → FM retained

Semantic subtask → FM chunk (~50-step/~1s @ 50 Hz) · NOT a classical planner · RTC companion

Club spine · one-liner

Open-world VLA via heterogeneous co-training + hierarchical inference:

FAST discrete pretrain → FM retained for continuous control.

Semantic subtask → ~50-step / ~1s FM chunk @ 50 Hz. NOT a classical planner.

RTC companion for async soft-inpaint. Blogs: “FAST replaced FM” = FALSE.

BEHAVIOR'26: $\pi 0.5$ + GR00T baselines · Oct 2 = $\pi 0.5$ -baseline + deltas.

Scientific Reviewer · strengths

► Unseen-scene protocol

Mock + 3 real kitchens/bedrooms · rubric partial credit · ~10 trials

► Multi-axis ablations

ME/CE/WD/VI/HL with coherent manip vs semantics stories

► Scaling curves

Locations 3→104 · $104 \approx$ test-home control under full co-train

► Hybrid recipe clarity

Fig.3 two-stage FAST→FM · attention separates discrete/continuous

► OpenPI bridge

Public weights make partial reproducibility real

Caveats · citation hygiene

Figure-only bars

Task-progress % often not numeric in text — PDF eyes required

Private MM / ME / CE / VI

Full open-world recipe irreproducible; use OpenPI proxies

Abbreviated scaling recipe

No full pretrain per location count — causal purity limited

Human HL oracle beaten

Surprising; possible interface mismatch / rater noise

KI clearer in follow-up

Stop-grad formalized in 2505.23705, not fully named here

Cite hybrid accurately

FAST pretrain + FM post-train/infer — never “FAST replaced FM”

FM vs FAST · say this aloud

Hybrid FAST discrete pretrain → FM retained for continuous control

Claim	Verdict
FAST replaced FM	MYTH
FAST pretrain + FM post-train	TRUE
FM head retained at inference	TRUE
OpenPI FM API for $\pi_{0.5}$	CLEAR
FAST is the deployed LL controller	FALSE

OpenPI ships FM head for $\pi_{0.5}$ · KI (2505.23705) formalizes stop-grad FAST+FM

Hybrid recipe · FAST discrete pretrain → FM retained

Bust the myth: FAST does NOT replace FM at deploy

① PRE-TRAIN · 280k steps

Discrete next-token VLA

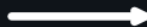
SigLIP 400M + Gemma ~2B

text · boxes · FAST action tokens · $\alpha=0$

FAST action tokens

scalable discrete pretrain signal

Mixture: MM · ME · CE · HL · WD



② POST-TRAIN / INFER · 80k

Joint NTP + flow matching $\alpha=10$

Action expert ~300M · FM

Ada-RMSNorm on τ · random init expert

Inference: AR subtask → 10 FM steps

~50-step / ~1s chunk @ 50 Hz PD

MM+ME+WD+VI · drop CE · FM RETAINED

NOT classical planner · same model for HL + LL · RTC = async soft-inpaint companion (2506.07339)

environments present persistent challenges (e.g., unfamiliar handles on drawers, or cabinets that are physically hard for the robot to open), some behaviors present challenges with partial observability (e.g., the robot arm occluding a spill that should be wiped), and in some cases the high-level sub-task inference is easily distracted (e.g., closing and opening a drawer multiple times while putting away items). Addressing these challenges with better co-training, transfer, and larger datasets is a promising direction for future work. Other future work directions could address the technical constraints of our method. While $\pi_{0.5}$ can perform a variety of behaviors to clean up kitchens and bedrooms, it processes relatively simple prompts. The complexity of the prompts that the model can accommodate is determined by the training data, and more complex preferences and instructions could be incorporated by producing more intricate and diverse annotations, either with human labelers or synthetically. The model also uses a relatively modest context, and incorporating richer context and memory could make the model significantly more capable in settings with more partial observability, such as tasks that require navigating between different rooms or remembering where objects are stored. More broadly, $\pi_{0.5}$ explores a particular combination of heterogeneous data sources, but the specific sources of data can be explored even more broadly. For instance, the ability of our system to learn from verbal instructions provides a powerful new supervision modality, and future work could explore this and other ways that people can provide robots with additional contextual knowledge. We hope that our work will serve as a foundation for a new generation of VLAs that exhibit broad generalization to diverse real-world

- scale manipulation platform for scalable and intelligent embodied systems. *arXiv preprint arXiv:2503.06669*, 2025.
- [2] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Daniel Ho, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Eric Jang, Rosario Jauregui Ruano, Kyle Jeffrey, Sally Jesmonth, Nikhil Joshi, Ryan Julian, Dmitry Kalashnikov, Yuheng Kuang, Kuang-Huei Lee, Sergey Levine, Yao Lu, Linda Luu, Carolina Parada, Peter Pastor, Jornell Quiambao, Kanishka Rao, Jarek Rettinghouse, Diego Reyes, Pierre Sermanet, Nicolas Sievers, Clayton Tan, Alexander Toshev, Vincent Vanhoucke, Fei Xia, Ted Xiao, Peng Xu, Sichun Xu, Mengyuan Yan, and Andy Zeng. Do as i can and not as i say: Grounding language in robotic affordances. In *arXiv preprint arXiv:2204.01691*, 2022.
- [3] Suneel Belkhale and Dorsa Sadigh. Minivla: A better vla with a smaller footprint, 2024. URL <https://github.com/Stanford-ILIAD/opensvla-mini>.
- [4] Suneel Belkhale, Tianli Ding, Ted Xiao, Pierre Sermanet, Quon Vuong, Jonathan Tompson, Yevgen Chebotar, Debiddatta Dwibedi, and Dorsa Sadigh. Rt-h: Action hierarchies using language, 2024. URL <https://arxiv.org/abs/2403.01823>.
- [5] Lucas Beyer, Andreas Steiner, André Susano Pinto, Alexander Kolesnikov, Xiao Wang, Daniel Salz, Maxim Neumann, Ibrahim Alabdulmohsin, Michael Tschannen, Emanuele Bugliarello, et al. Paligemma: A versatile 3b vlm for transfer. *arXiv preprint arXiv:2407.07726*, 2024.
- [6] Homanga Bharadhwaj, Jay Vakil, Mohit Sharma, Abhinav Gupta, Shubham Tulsiani, and Vikash Kumar

[10]

[11]

[12]

Verdict

Credible advance on open-world VLA generalization.

Strong system paper. Hybrid FAST→FM is scientifically central
and must be cited accurately.

“First” new-home claim is bold but scoped to learning-enabled VLAs of this complexity.

Novelty = recipe + hierarchy + eval — more than new FM math.

Scientific Reviewer · close

Credible open-world VLA advance. Cite hybrid FAST→FM accurately.

Caveats: figure-only bars · private MM · human-HL · KI clearer in follow-up

Q&A · Sep 30 · $\pi_{0.5}$ / 2504.16054

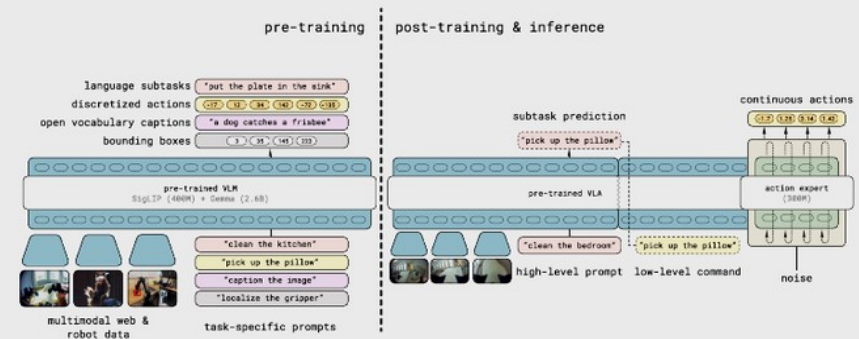


Fig. 3: **Model overview.** $\pi_{0.5}$ is trained in two stages. First, a pre-training stage combines all of the different data sources to produce an initial VLA with discrete tokens. This stage uses data from diverse robotic platforms, high-level semantic action prediction, and data from the web. Robotic data uses the FAST action tokenizer to represent actions as discrete tokens [64]. Second, a post-training stage specializes the model for low-level and high-level inferences for mobile manipulation, leveraging the most task-relevant data, including verbal instructions from human supervisors. This stage uses flow matching to represent the action distribution, enabling efficient real-time inference and the ability to represent fine-grained continuous action sequences. At inference time, the model first infers a high-level subtask, and then predicts the actions based on this subtask.

primitives, such as picking up objects, methods that allow for task-specific assumptions (e.g., grasp prediction, or incorporating model-based planning and control) have been shown to generalize broadly, even to entirely new homes [40, 20, 60, 56, 29]. However, such methods do not readily generalize to the full range of possible tasks that a generalist robot might need to perform. More recently, large-scale datasets collected across many domains [41, 68, 63, 67, 14, 49] have been shown to enable generalization of simple but end-to-end learned tasks to new environments [33, 31, 67, 69, 26, 49, 28, 64]. However, the tasks in these demonstrations are still relatively simple, typically less than a minute in length and often with relatively low success rates. We show that $\pi_{0.5}$ can perform long, multi-stage tasks, such as putting all of the dishes in the sink or picking all of the clothing off the floor of a new bedroom, while generalizing to entirely new homes.

III. PRELIMINARIES

Vision-language-action models (VLAs) are typically trained via imitation learning on diverse robot demonstration datasets $\mathcal{D}$, by maximizing the log-likelihood of an action $\mathbf{a}_t$ (or, more generally, an action *chunk* $\mathbf{a}_{t:t+H}$) given an observation $\mathbf{o}_t$ and a natural language task instruction ℓ : $\max_{\theta} \mathbb{E}_{(\mathbf{a}_{t:t+H}, \mathbf{o}_t, \ell) \sim \mathcal{D}} \log (\pi_{\theta}(\mathbf{a}_{t:t+H} | \mathbf{o}_t, \ell))$. The observation typically contains one or more images $\mathbf{I}_1^t, \dots, \mathbf{I}_n^t$ and proprioceptive state $\mathbf{q}_t$, which captures the position of the robot's joints. VLA architectures follow the design of modern language and vision-language models, with modality-specific tokenizers that map inputs and outputs to discrete ("hard") or continuous ("soft") token representations, and a large, autoregressive transformer backbone that is trained to map from

input to output tokens. The weights of these models are initialized from pre-trained vision-language models. By encoding policy inputs and outputs into tokenized representations, the imitation learning problem described above can be cast as a simple next-token-prediction problem over a sequence of observation, instruction and action tokens, and we can leverage the scalable tools of modern machine learning to optimize it. In practice, the choice of tokenizers for image and text inputs follows those of modern vision-language models. For actions, prior work has developed effective, compression-based tokenization approaches [64], which we use in this work during pretraining. A number of recent VLA models have also proposed to represent the action distribution via diffusion [55, 84, 52] or flow matching [8], providing a more expressive representation over continuous-valued action chunks. During the post-training phase of our model, we will build on the design of the π_0 model [8], which represents the action distribution via flow matching. In this design, the tokens corresponding to actions receive the partially denoised actions from the previous step of flow matching as input, and output the flow matching vector field. These tokens also use a different set of model weights, which we refer to as an "action expert," analogously to a mixture of experts architecture. This action expert can specialize to flow matching-based action generation, and can be significantly smaller than the rest of the LLM backbone.

IV. THE $\pi_{0.5}$ MODEL AND TRAINING RECIPE

We provide an overview of the $\pi_{0.5}$ model and training recipe in Figure 3. The model weights are initialized from a standard VLM trained on data from the web, and training then